THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS UNIT TEST SET C

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The 11th triangular number is: 1

(a) 77 **(b)** 64 **(c)** 66 **(d)** 67

2. The number of sides in the shape sequence of regular polygons starting with the triangle is: 1

(a) 1, 3, 6, 10, … **(b)** 3, 4, 5, 6, … **(c)** 2, 4, 6, 8, … **(d)** 3, 6, 9, 12, …

3. An angle of 135° is: 1

(a) obtuse **(b)** right **(c)** acute **(d)** straight

4. A point is shown in a drawing by a: 1

(a) line **(b)** dot **(c)** square **(d)** arc

5. Which is the largest 5-digit number? 1

(a) 10000 **(b)** 99999 **(c)** 98765 **(d)** 99990

6. Starting at 6, the Collatz sequence is 6, 3, 10, 5, 16, 8, 4, 2, 1. The number of steps is: 1

(a) 9 **(b)** 10 **(c)** 8 **(d)** 6

7. Rounding 6743 to the nearest thousand gives: 1

(a) 6000 **(b)** 6700 **(c)** 6800 **(d)** 7000

8. In a pictograph, the key tells us: 1

(a) how many items one symbol stands for **(b)** the colour
(c) the date **(d)** the title

9. The data 3, 5, 3, 7, 3 has the number 3 with a frequency of: 1

(a) 1 **(b)** 3 **(c)** 5 **(d)** 2

10. In a pictograph where one symbol stands for 16, how many symbols are needed for 96? 1

(a) 7 symbols **(b)** 6 symbols **(c)** 5 symbols **(d)** 8 symbols

11. The number 1 is: 1

(a) neither prime nor composite **(b)** even
(c) prime **(d)** composite

12. The factors of a number are always: 1

(a) greater than the number **(b)** primes only
(c) less than or equal to the number **(d)** odd only

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SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. What is the 7th term of the cube-number sequence? 2

14. What are collinear points? Give an example. 2

15. Write the Collatz sequence starting from 5. 2

16. What is a frequency table? 2

17. Find the LCM of 4 and 10. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. The number of lines in complete graphs of 2, 3, 4, 5 points are 1, 3, 6, 10. What sequence is this and what will be the next term? 3

19. Draw and name three points that are not collinear. How many lines can join them in pairs? 3

20. Make 5 different 4-digit palindromes and find their sums with their reverses. 3

21. Find the factors of 36 and state how many there are. 3

22. What patterns are seen in the sequence of cube numbers? 3

OR

Find the sum $1 + 2 + 3 + \cdots + 10$ using a pattern. Explain. 3

23. Find the angle between the hands of a clock at 4 o'clock and 8 o'clock. 3

OR

Describe the steps to measure an angle with a protractor. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) Define point, line segment, ray and line. 5
(ii) Draw each.
(iii) Mention the number of end points of each.

OR

Case study: A ladder rests against a wall making 70° with the ground. 5
(i) What is the angle between the ladder and the wall if the wall is vertical?
(ii) Name the types of the two angles.
(iii) What would happen if the angle were 90°?
(iv) Which tool measures it?

25. **Case study:** A pictograph shows the books borrowed: Mon ●●●, Tue ●●●●●, Wed ●●, with ● = 12 books. 5
(i) Books borrowed each day?
(ii) The total?
(iii) On which day was the least?
(iv) How many more on Tue than on Wed?

OR

(i) Describe how to draw a pictograph. 5
(ii) Draw a pictograph for 20, 35, 15 (1 symbol = 5).
(iii) Which item is the lowest?

END OF QUESTION PAPER — Total: 50 Marks